Code-switching seems to act as a prominence cue for early Spanish-English bilinguals, but we do not observe a comparable pattern in L2 German-English learners.

Prominence-signaling functions of code-switching in L2 learners and early bilinguals

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References, Poster, Preregistration, Data, Scripts

German-English

Spanish-English

Why code-switch?

Code-switching: changing languages in the middle of a conversation or sentence

Why do bilinguals code-switch in production?

- Identity marker
- Communicative efficiency (lexical gaps; order of information; quotations)
- Signaling of information that is unusual
  - Infrequent words (Tomić and Valdés Kroff, 2021)
  - Emotional information (Tomić and Valdés Kroff, 2023)
  - Unexpected information (Myslín and Levy, 2015) → **Here**

Participants

- Gen-Z (18-35 years old)
- **German-English version (Germany):**
  - 120 L1 German L2 English speakers (AoA German < 4 years; AoA English < 12 (mainly at school); Lextale English > 60%)
- **Spanish-English version (USA):**
  - 122 early bilinguals (AoA Spanish < 6 years; AoA English < 12 years; Lextale Spanish > 55%)

Including: language background data, attitudes towards language mixing, frequency of code-mixing, proficiency in each language

Preregistered hypothesis

Code-switching is a signal to the reader/listener that unusual information is coming up

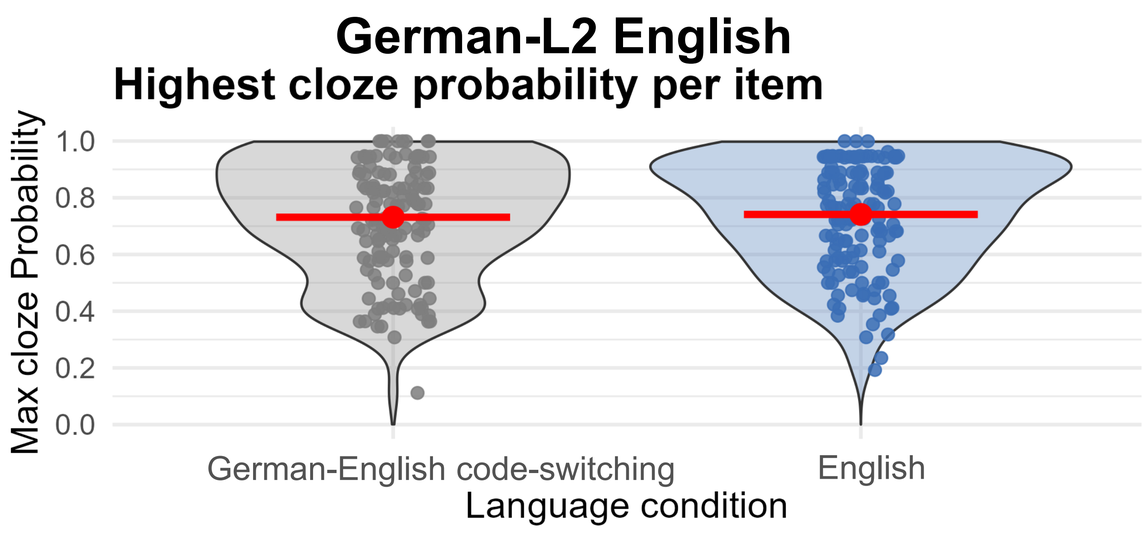
This predicts:

- Cloze probability of best completion will be smaller in the code-switched than unilingual condition
- Type-token ratio will be larger in the code-switched than unilingual condition

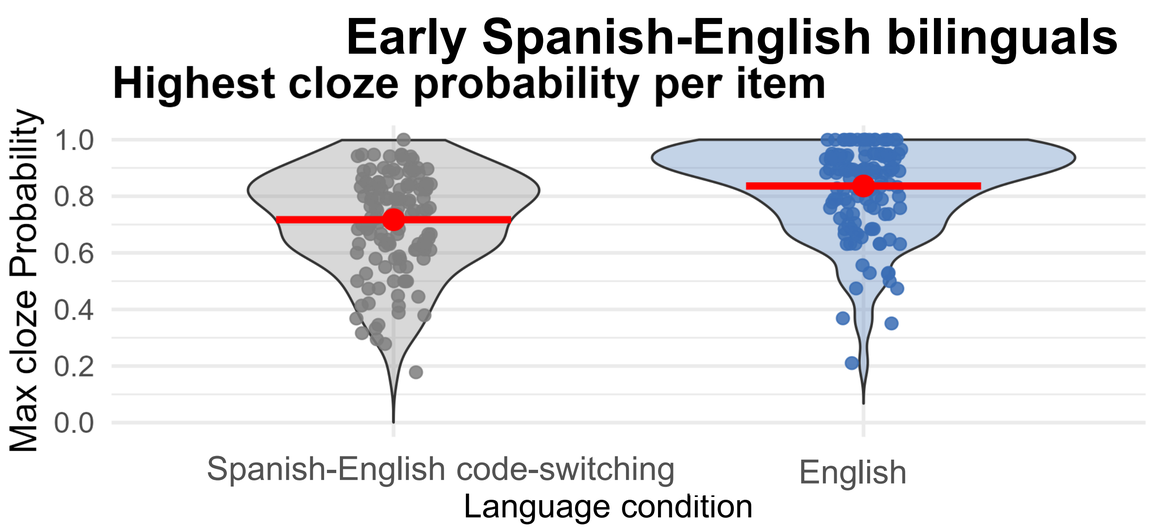
Design

- Cloze-questionnaire sentence completion task
- Unilingual English and code-switches ending in English
- Instructed to complete target word in English in either condition
- Highly constraining contexts (133 items analyzed in Spanish-English version, 144 items analyzed in German-English version; based on Federmeier et al. 2007)
- Each item version completed by 15-27 participants

Unilingual English	
They raised pigs **on their** _____	farm (expected completion) roof (unexpected completion)
Code-switch from German to English	**Code-switch from Spanish to English**
Sie züchteten Schweine **on their** _____	Ellos criaban cerdos **on their** _____



No significant differences between code-switched German-English and unilingual English conditions: b = 0.01, p = 0.38, SE = 0.01



Significant differences between code-switched Spanish-English and unilingual English conditions: b= 0.62, p < .0001, SE = 0.01

Discussion

- Evidence that Code-switching cues less expected information in early Spanish-English bilinguals, but not in German-L2 English speakers:
  - German-L2 English, L2: Code-switching does not affect cloze probabilities
  - Early Spanish-English: Code-switching leads to lower cloze probabilities
  - The patterns for type-token ratios mirror those for cloze probabilities
- Individual differences:
  - Lower proficiency in Spanish (early bilinguals) or English (German group) → less likely to produce the highest cloze word in **both** code-switch and unilingual English conditions. So: lower cloze values in code-switching vs. unilingual conditions cannot be fully explained by lower proficiency

Next steps

→ Do bilinguals use code-switching as a signaling cue **during processing**?

- Investigation via ERP-study with materials from the cloze study
  - If code-switching is a cue for unexpected information: N400 / post-N400 positivity will be smaller in the code-switch than in the unilingual condition
  - If code-switching is just hard: N400 will be overall larger in the code-switching than in the unilingual condition
- Further exploration of the type of bilingualism and code-switching by adding more combinations: Spanish-German, Chinese-English, Turkish-German

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